

User agent

Text request

 Return amount in May?

ask “amount”

then
writes SQL

SQL request

```
SELECT SUM(sr_return_amt)
FROM store_returns
JOIN date_dim ON ...
WHERE sr_status = 'done'
AND d_moy = 5 AND ...
```

declares amount v2

MAVRA query service

1 Match name
name or alias

2 Data versions
last check → now

3 Validate (cache first)
look up each rule; validate misses

4 Return
only if valid

return amount ✓

return rate —

revenue —

then now

store_returns 14 15

date_dim 3 3

changed: recheck

one row/return cache hit: reuse ✓

date key unique cache hit: reuse ✓

dates complete miss: check, cache ✓

amount v2, valid

store_returns

SUM(sr_return_amt)

sr_status='done'

Validation cache

Results are cached per (rule, data version).
Same rule, same data version: every definition
and every agent shares one result. Only a new
data version needs a new check.

rule	data version		written by	used in
one row/return	store_returns 15	✓	return rate	3
date key unique	date_dim 3	✓	earlier	3 6
dates complete	store_returns 15	✓	3	
one row/return	store_returns 16	✓	6	
dates complete	store_returns 16	✓	6	

5 Check version
is the named v2 usable?

6 Snapshot check
do the rules hold on this snapshot?

7 Run
on the same snapshot

✓ v2 exists, not retired

✓ v2 is the latest

✓ SQL keeps the filter

✗ naming v1: rejected

snapshot sees store_returns 16

one row/return miss: check here ✓

date key unique hit: reuse ✓

dates complete miss: check here ✓

run the SQL

result 3,294,349.93

a failed rule

rejects the SQL